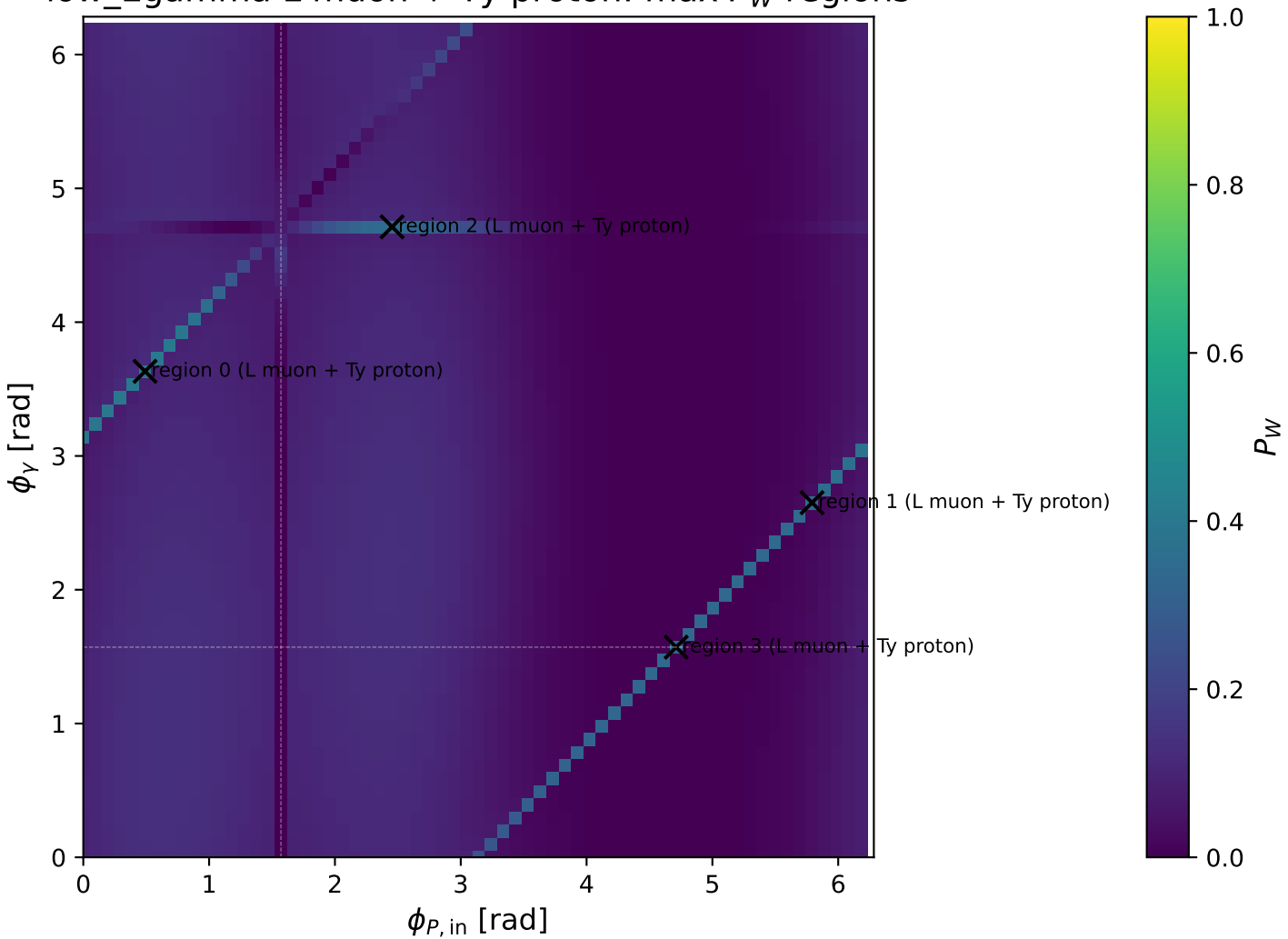
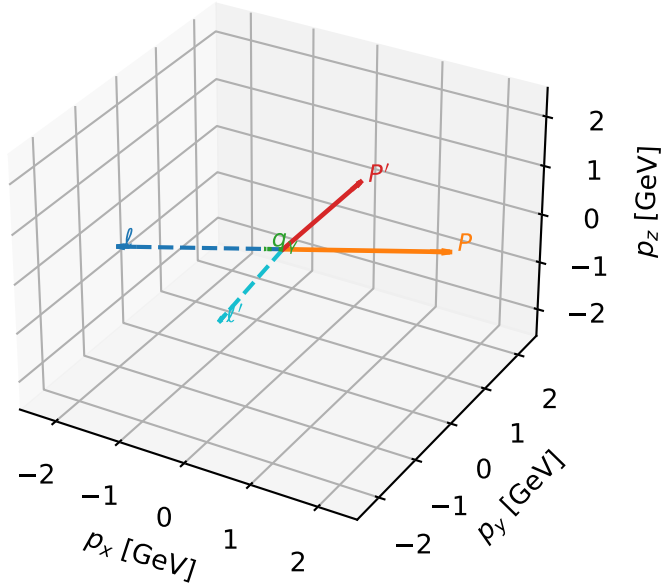


low_Egamma L muon + Ty proton: max P_W regions



L muon + Ty proton characteristic max P_W configuration

3D momenta

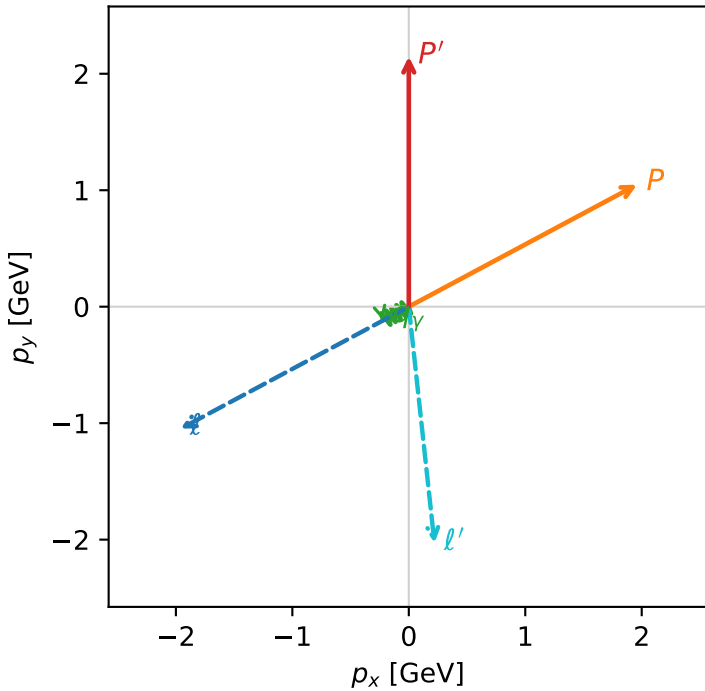


w_purity_low egamma_lty_region_0 P_W=0.408856
 region: low_Egamma_LTy_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_\mu\rangle \otimes |T_{y,p}\rangle$

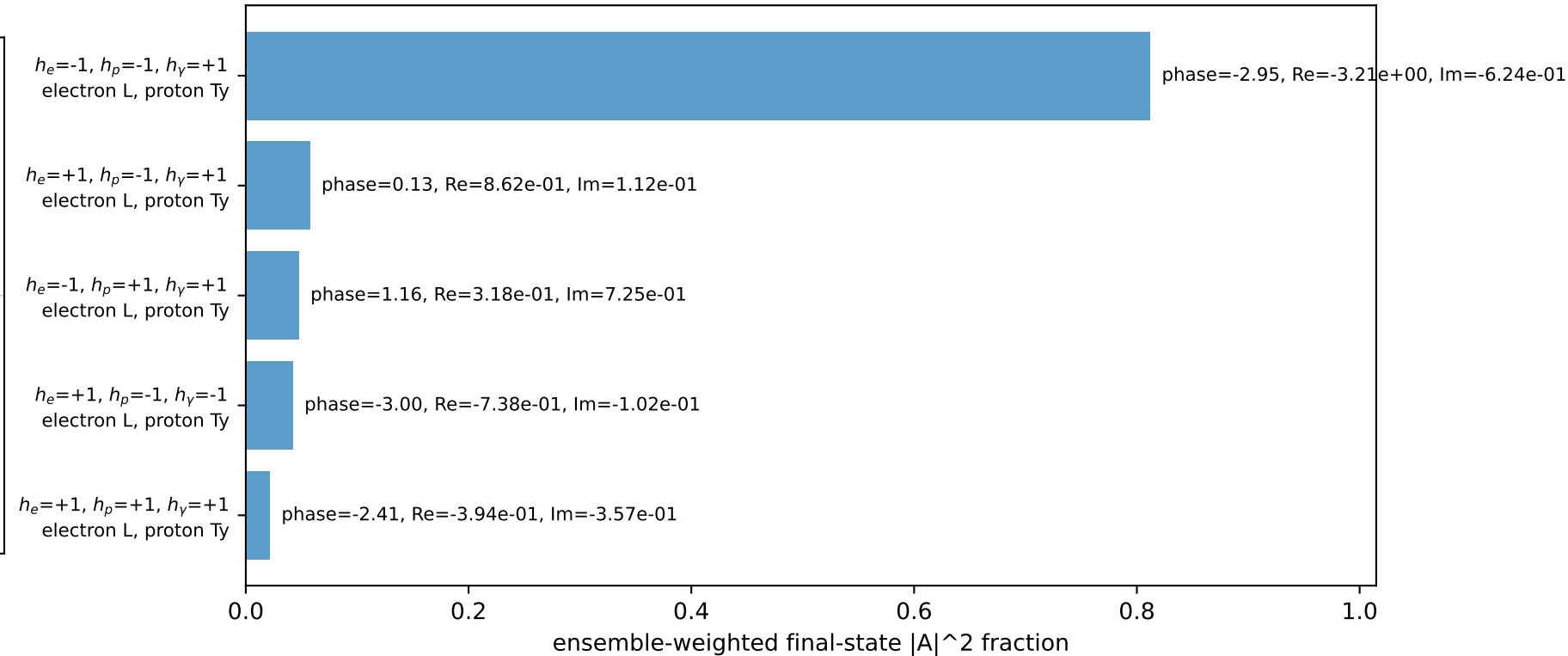
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.14791$
 $\theta_{in}=1.57079$, $\phi_l=3.63247$, $\phi_P=0.490874$
 $E_Y=0.25$, $\phi_Y=3.63247$
 $Q^2=5.68893$, $x_B=0.772209$, $t=-5.04908$, $W^2=2.558$, $y=0.357195$
 $\theta(l', \gamma)=68.0735$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -1.9606$ $p_y= -1.0480$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 1.9606$ $p_y= 1.0480$ $p_z= 0.0000$
 l' $E= 2.0447$ $p_x= 0.2205$ $p_y= -2.0301$ $p_z= 0.0000$
 P' $E= 2.3438$ $p_x= 0.0000$ $p_y= 2.1479$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= -0.2205$ $p_y= -0.1178$ $p_z= 0.0000$

Transverse plane

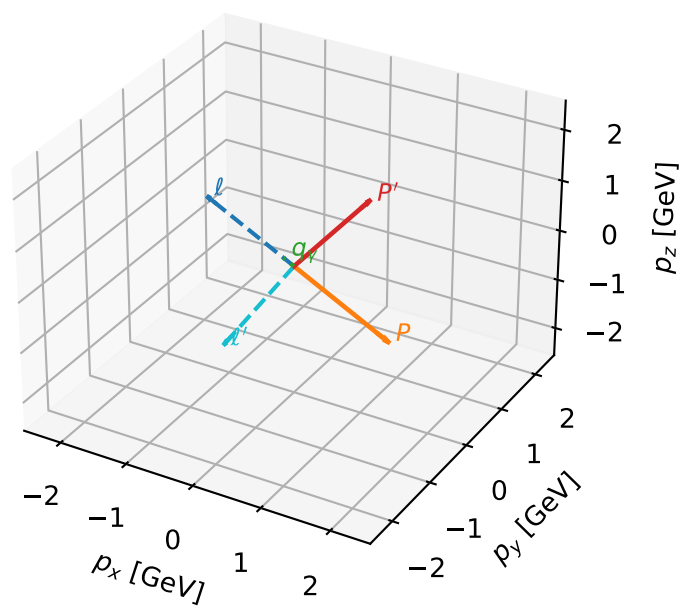


Leading initial-ensemble/final-state components (N=5, retained=98.1%)



L muon + Ty proton characteristic max P_W configuration

3D momenta

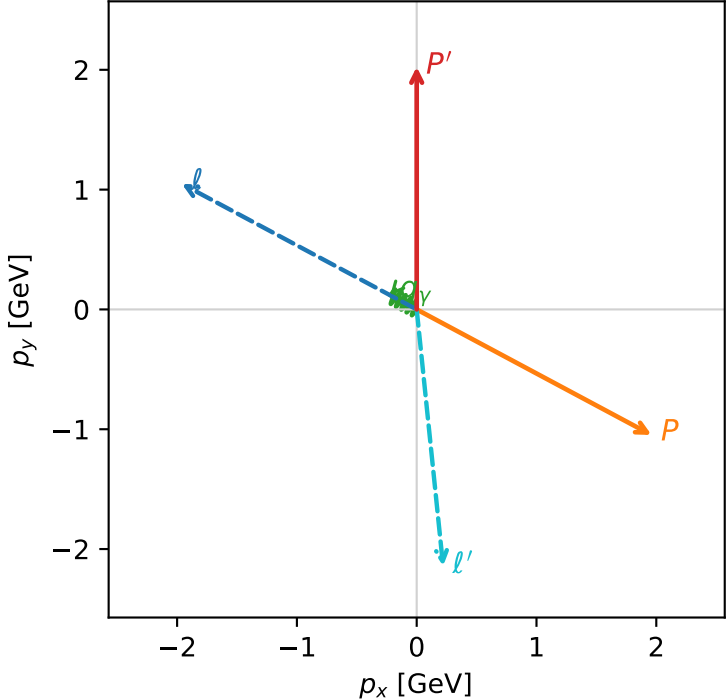


w_purity_low_egamma_lty_region_1 P_W=0.347803
region: low_Egamma_LTy_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_\mu\rangle \otimes |T_{\gamma,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.02504$
 $\theta_{in}=1.57079$, $\phi_\ell=2.65072$, $\phi_P=5.79231$
 $E_\gamma=0.25$, $\phi_\gamma=2.65072$
 $Q^2=14.934$, $x_B=0.958994$, $t=-13.2545$, $W^2=1.51841$, $y=0.755038$
 $\theta(\ell', \gamma)=123.999$ deg

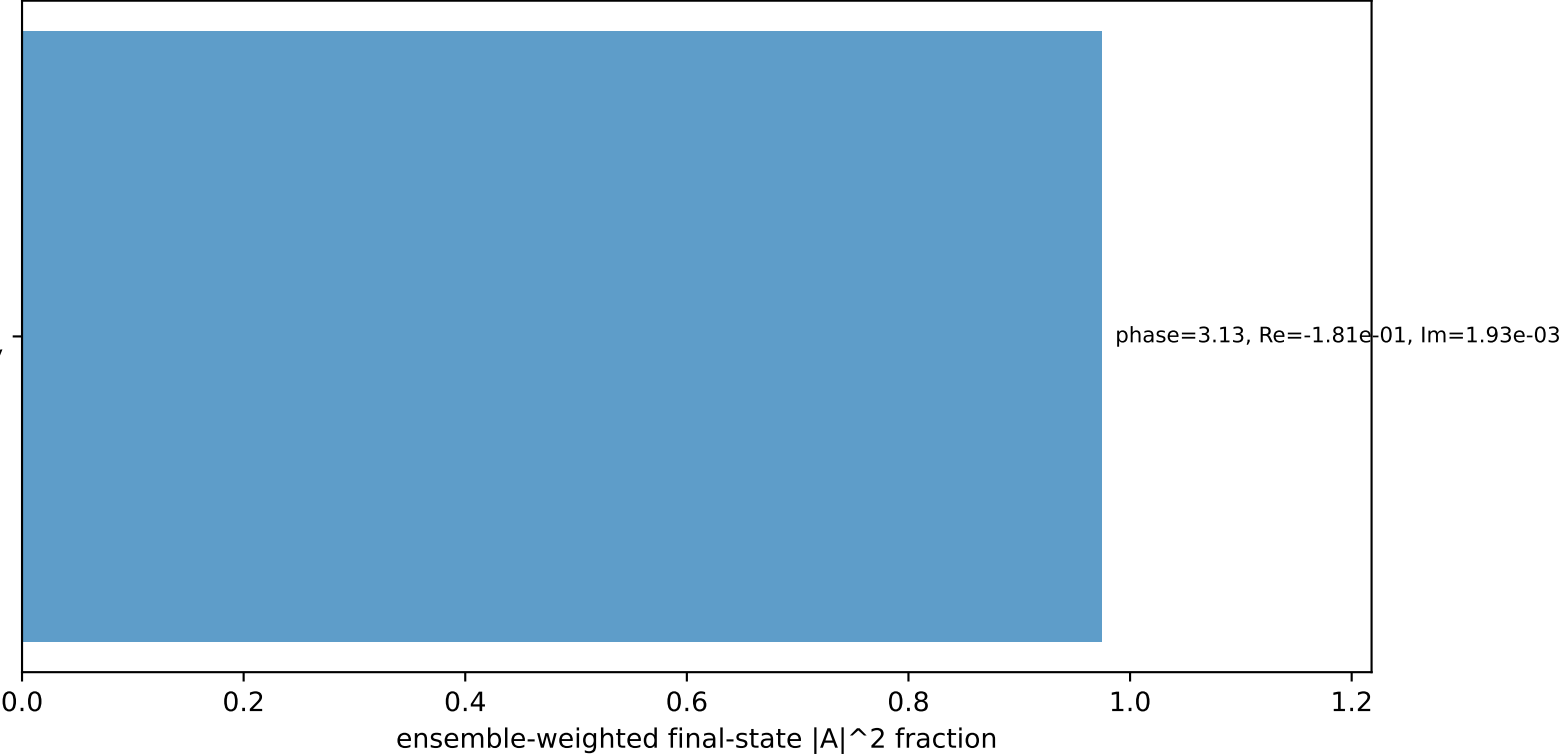
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2256$ $p_x= -1.9606$ $p_y= 1.0480$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 1.9606$ $p_y= -1.0480$ $p_z= 0.0000$
 ℓ' $E= 2.1568$ $p_x= 0.2205$ $p_y= -2.1429$ $p_z= 0.0000$
 P' $E= 2.2317$ $p_x= 0.0000$ $p_y= 2.0250$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.2205$ $p_y= 0.1178$ $p_z= 0.0000$

Transverse plane



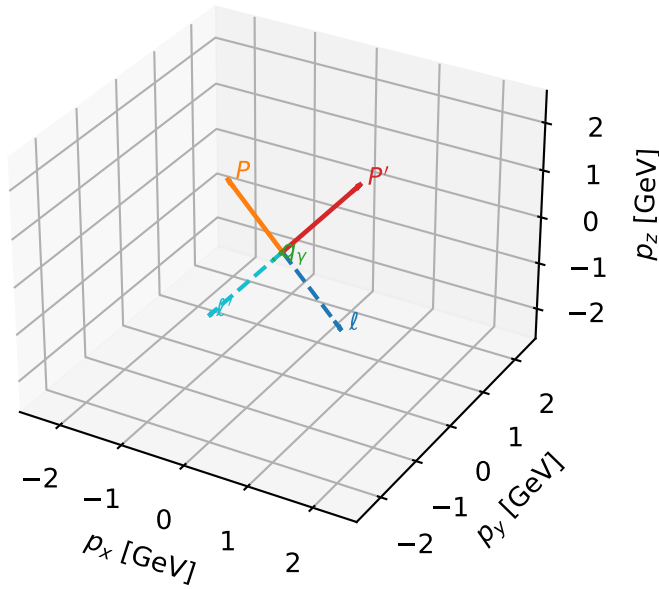
$h_e=-1, h_p=-1, h_\gamma=+1$
electron L, proton Ty

Leading initial-ensemble/final-state components (N=1, retained=97.4%)



L muon + Ty proton characteristic max P_W configuration

3D momenta

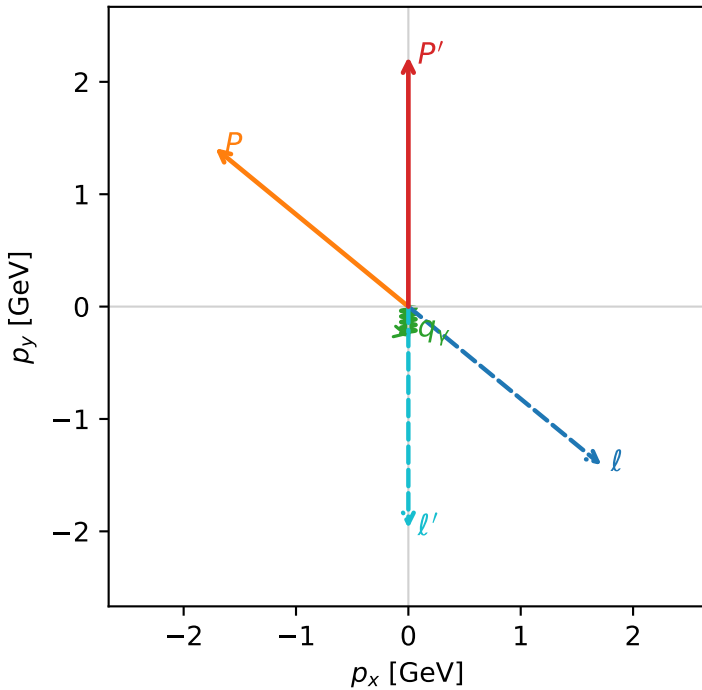


w_purity_low egamma lty_region 2 P_W=0.342044
 region: low_Egamma_LTy_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $|L_\mu\rangle \otimes |T_{y,p}\rangle$

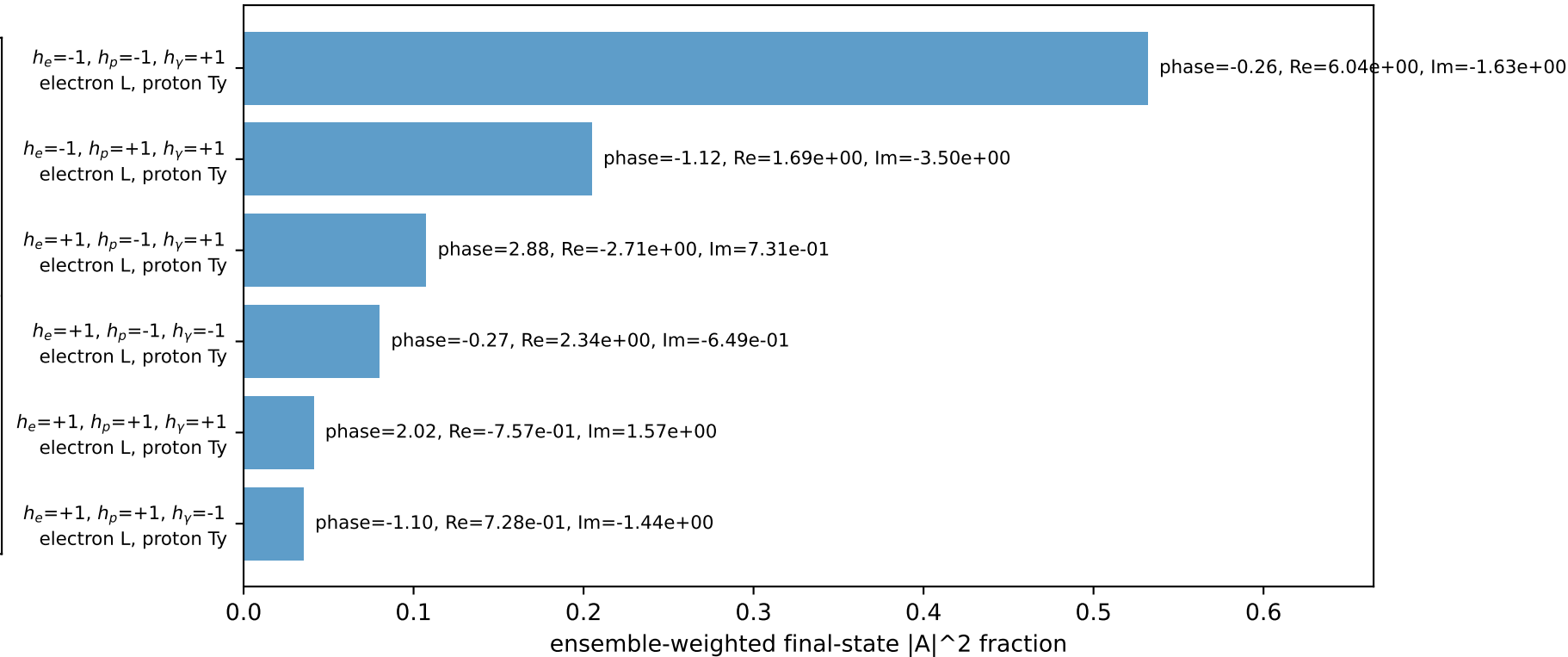
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.22295$
 $\theta_{in}=1.57079$, $\phi_l=5.59596$, $\phi_p=2.45437$
 $E_Y=0.25$, $\phi_Y=4.71239$
 $Q^2=3.20732$, $x_B=0.580493$, $t=-3.61355$, $W^2=3.19769$, $y=0.267889$
 $\theta(l', \gamma)=0$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 1.7185$ $p_y= -1.4103$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -1.7185$ $p_y= 1.4103$ $p_z= 0.0000$
 l' $E= 1.9758$ $p_x= 0.0000$ $p_y= -1.9729$ $p_z= 0.0000$
 P' $E= 2.4127$ $p_x= 0.0000$ $p_y= 2.2229$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= -0.0000$ $p_y= -0.2500$ $p_z= 0.0000$

Transverse plane

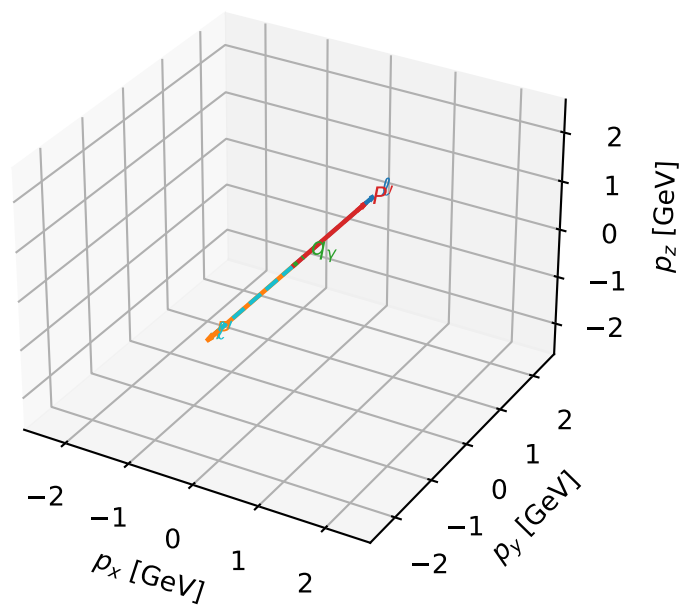


Leading initial-ensemble/final-state components (N=6, retained=100.0%)



L muon + Ty proton characteristic max P_W configuration

3D momenta

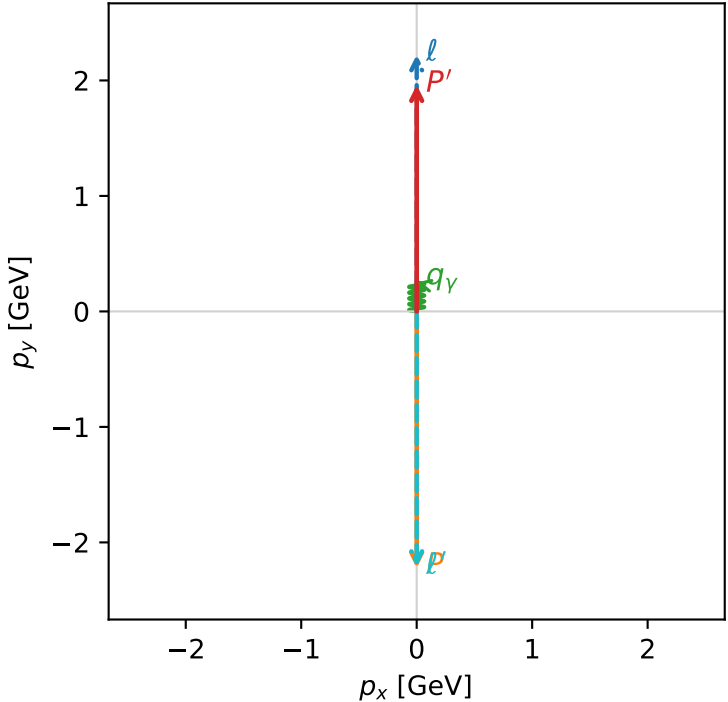


w_purity_low_egamma_lty_region_3 P_W=0.333333
region: low_Egamma_LTy_region_3
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_\mu\rangle \otimes |T_{Y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.96163$
 $\theta_{in}=1.57079$, $\phi_l=1.5708$, $\phi_p=4.71239$
 $E_Y=0.25$, $\phi_Y=1.5708$
 $Q^2=19.6668$, $x_B=0.994621$, $t=-17.4552$, $W^2=0.986208$, $y=0.958708$
 $\theta(l', \gamma)=180$ deg

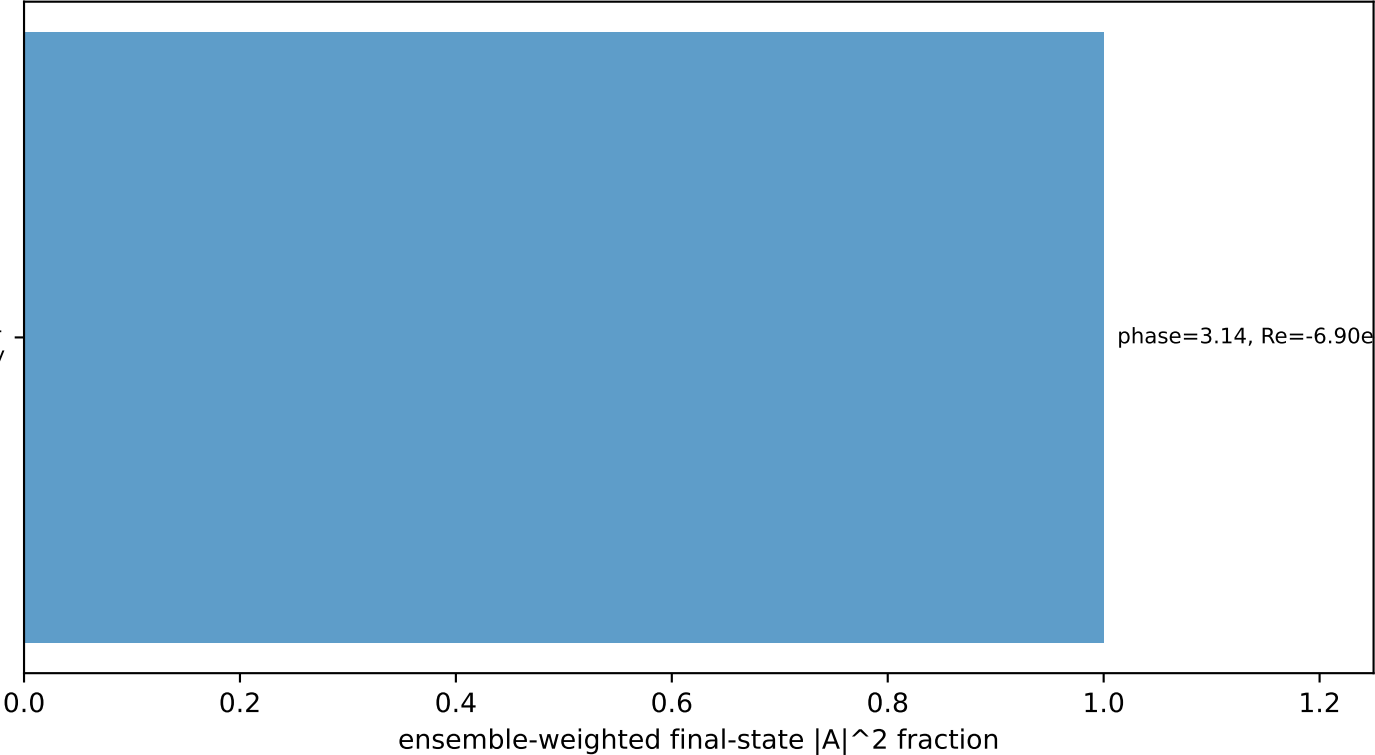
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 0.0000$ $p_y= 2.2231$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -0.0000$ $p_y= -2.2231$ $p_z= 0.0000$
 l' $E= 2.2142$ $p_x= -0.0000$ $p_y= -2.2116$ $p_z= 0.0000$
 P' $E= 2.1744$ $p_x= 0.0000$ $p_y= 1.9616$ $p_z= 0.0000$
 q_Y $E= 0.2500$ $p_x= 0.0000$ $p_y= 0.2500$ $p_z= 0.0000$

Transverse plane



$h_e=-1$, $h_p=-1$, $h_Y=+1$
electron L, proton Ty

Leading initial-ensemble/final-state components (N=1, retained=100.0%)



phase=3.14, Re=-6.90e-02, Im=2.55e-19